How long does a scrambled quantum memory remember?

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Scrambling spreads quantum information over high-weight Pauli operators, which local noise damps exponentially, so scrambled encodings are expected to be fragile. We show they survive up to a sharp threshold. In an idling memory, states from any state 2-design remain distinguishable until the Pauli error entropy reaches one bit per qubit. A random codebook of such states carries classical information at exactly the rate $1 - H$, and beyond the threshold a single noise layer forces exponential error-mitigation cost. Purity bounds overestimate the memory lifetime almost twofold.

Introduction.—Quantum information is rarely stored in simple states. A quantum memory, a register waiting between gates, or the output of a generic circuit holds a delocalized, scrambled state, modeled by a state 2-design: an ensemble, such as random stabilizer states, whose first two moments match Haar-random states [1]. While it waits, every qubit decoheres. How long does the stored information last?

The standard picture suggests it does not last long. A scrambled state places almost all of its Pauli weight on operators that act on a constant fraction of the qubits, and local depolarizing noise of strength r multiplies a weight-w operator by $(1-r)^w$. At $r = 0.2$ and $n = 100$ qubits, the surviving Pauli weight of a 2-design state is about 10^{-14}. The same mechanism underlies the exponential sample cost of quantum error mitigation (QEM). Takagi, Tajima, and Gu [2] and Tsubouchi, Sagawa, and Yoshioka [3] proved that this cost grows exponentially with circuit depth for any noise strength; related analyses appear in Refs. [4, 5]. Quek *et al.* [6] showed that random circuits reach this regime at very shallow depth, using purity to control the decay.

Pauli weight, however, is not information. We show that a scrambled memory keeps its information until a sharp threshold, fixed by the entropy of the single-qubit errors, and we identify three consequences. First, a memory with coherence times $T_1 = T_2$ retains information for $0.291\,T_1$, whereas purity certifies loss only after $0.549\,T_1$. Second, a codebook of scrambled states transmits classical information at exactly the hashing rate $1 - H$. Third, beyond the threshold, a single layer of noise already forces QEM to use exponentially many samples. For biased noise and pure dephasing, the gap to purity-based estimates widens qualitatively.

All three follow from one quantity. Two pure states ψ_1, ψ_2 are drawn independently from a state 2-design and stored under the same Pauli channel $\mathcal{N}$, with no further randomization. With equal priors, the optimal probability of identifying the stored state is $(1 + \overline{T})/2$, where

$$\overline{T} = \mathbb{E}_{\psi_1,\psi_2} \tfrac{1}{2} \|\mathcal{N}(\psi_1) - \mathcal{N}(\psi_2)\|_1. \tag{1}$$

The absence of randomization after the noise is essential. Averaging $U^\dagger \mathcal{N}(U \cdot U^\dagger) U$ over a unitary 2-design turns any Pauli channel into global depolarization, with parameter $\theta = (d^2 p_I - 1)/(d^2 - 1)$ [7], which decays exponentially in n for every $r > 0$. At $n = 8$ and $r = 0.2$, $\overline{T} = 0.736$ without the twirl, against about 0.272 with it [Fig. 1(a)].

Ibarrondo and Stilck França [8] and Liu *et al.* [9] bounded $\overline{T}$ for local depolarizing noise, finding near-perfect distinguishability below $r \approx 0.25$ and decay above $r \approx 0.42$, with the interval between them open. Sommers *et al.* [10] found the hashing transition, with an $n^{-1/2}$ window, in the coherent information of random codes. The bound below closes the interval and holds for every system size.

Information-spectrum bracket.—Let $d = 2^n$, $\mathcal{P}_n = \{I, X, Y, Z\}^{\otimes n}$, and

$$\mathcal{N}_p(\rho) = \sum_{P \in \mathcal{P}_n} p_P P \rho P, \qquad s(P) = -\log_2 p_P, \tag{2}$$

with p an arbitrary distribution, so errors may be correlated, and with the surprisal s distributed as $P \sim p$. Logarithms are base two. Each $\psi_i = |\psi_i\rangle\langle\psi_i|$ is an n-qubit pure state, generally entangled, drawn independently from an exact state 2-design on the full space,

$$\mathbb{E}\psi_i = I/d, \qquad \mathbb{E}\psi_i^{\otimes 2} = (I + F)/[d(d+1)], \tag{3}$$

where F swaps the two copies; the two ensembles may differ.

Theorem 1. *For every Pauli distribution p and every real a,*

$$\mathbb{P}[s < n - a] - 2^{-a} \le \overline{T} \le \mathbb{P}[s < n + a] + 2^{-a/2}/\sqrt{2}. \tag{4}$$

More generally, for any set $\mathcal{G} \subseteq \mathcal{P}_n$ and any partition $\{\mathcal{C}_j\}$ of $\mathcal{P}_n$, with $m_j = \sum_{\mathcal{C}_j} p_P$ and $v_j = \sum_{\mathcal{C}_j} p_P^2$,

$$\overline{T} \ge p(\mathcal{G}) - |\mathcal{G}|/d, \tag{5}$$

$$\overline{T} \le \sum_j \min\left\{ m_j, \sqrt{\frac{d^2 v_j - m_j^2}{2(d+1)}} \right\}. \tag{6}$$

Proof. Fix ψ_1 and let Π project onto the span of $\{P|\psi_1\rangle : P \in \mathcal{G}\}$. For each realization, $\frac{1}{2}\|\mathcal{N}_p(\psi_1) - \mathcal{N}_p(\psi_2)\|_1 \ge \operatorname{tr}\Pi[\mathcal{N}_p(\psi_1) - \mathcal{N}_p(\psi_2)]$. The first term is at least $p(\mathcal{G})$. Averaged over the independent ψ_2, the second is $\operatorname{tr}\Pi/d \le$

$|\mathcal{G}|/d$, using only $\mathbb{E}\psi_2 = I/d$. Averaging over ψ_1 gives (5), the typical-image argument of Ref. [8] applied to an arbitrary set of Pauli errors. For the upper bound let $\Delta = \psi_1 - \psi_2$ and $\mathcal{N}_j(\rho) = \sum_{\mathcal{C}_j} p_P P\rho P$. Pauli orthogonality and (3) give

$$\mathbb{E}\,\mathrm{tr}\,\mathcal{N}_j(\psi_i)^2 = \frac{dv_j + m_j^2}{d+1}, \qquad \mathbb{E}\,\mathrm{tr}\,\mathcal{N}_j(\psi_1)\mathcal{N}_j(\psi_2) = \frac{m_j^2}{d},$$
$$\mathbb{E}\,\mathrm{tr}\,\mathcal{N}_j(\Delta)^2 = \frac{2d}{d+1}v_j - \frac{2m_j^2}{d(d+1)}. \tag{7}$$

Jensen's inequality and $\|A\|_1 \le \sqrt{d}\|A\|_2$ bound $\mathbb{E}\frac{1}{2}\|\mathcal{N}_j(\Delta)\|_1$ by the square root in (6), and $\|\Delta\|_1 \le 2$ bounds it by m_j. For (4), take $\mathcal{G} = \{s < n - a\}$, which has fewer than $d2^{-a}$ elements, and split the partition at $s = n + a$, where the tail has $v \le 2^{-n-a}$. □

The theorem has a simple reading. The error images $P|\psi_1\rangle$ of a scrambled state are nearly orthogonal, and a d-dimensional space holds at most d of them. Errors with $p_P > 1/d$, i.e., $s(P) < n$, number fewer than d, so their images fit into the space, and the projector onto these images separates ψ_1 from an independent ψ_2. The remaining errors are too numerous to fit, and their contributions average out. Distinguishability is therefore pinned between the probabilities of a likely error at surprisal $n \mp a$, up to exponentially small corrections. We checked this argument in Lean 4 with Mathlib [11, 12]: from Eq. (5) and the two-class case of Eq. (6) as hypotheses, the formal proof derives Eq. (4) and Corollary 1 below, and it proves the counting and moment algebra in full [13].

The trivial partition, $\mathcal{C} = \mathcal{P}_n$, reduces (6) to the purity of the noisy output, $\overline{T} \le [d\sum_P p_P^2/2]^{1/2}$, the quantity that controls the bounds of Ref. [6]. Purity is loose because it places likely and unlikely errors in the same second moment. Separating them at surprisal n removes this looseness.

Sharp threshold.—For i.i.d. noise, $p = \boldsymbol{p}^{\otimes n}$, with single-qubit distribution $\boldsymbol{p} = (p_0, p_x, p_y, p_z)$, entropy $H = \mathbb{E}[-\log_2 p_i]$, and variance $\sigma^2 = \mathrm{Var}[-\log_2 p_i]$. The surprisal is then a sum of n independent terms with mean nH. Taking $a = \epsilon n$ in (4) and applying a large-deviation bound [14] gives

$$\overline{T} = \begin{cases} 1 - e^{-\Omega(n)}, & H < 1, \\ e^{-\Omega(n)}, & H > 1. \end{cases} \tag{8}$$

The threshold $H = 1$ is the hashing point, at which the coherent information $1 - H$ of the channel with maximally mixed input vanishes [15, 16]. For depolarizing noise $\mathcal{N}_r(\rho) = (1-r)\rho + rI/2$, it lies at $r_c = 0.252386\ldots$, whereas purity certifies loss only for $r > 1 - 1/\sqrt{3} = 0.423$. Above threshold the decay rate is at least the universal$_2$ privacy-amplification exponent [17, 18]; near r_c both upper and lower rates equal $2.64\,(r - r_c)^2$ to leading order [13].

TABLE I. Retention time of a scrambled memory, in units of T_1, from the hashing condition $H = 1$ and from the purity bound, for Pauli-twirled idling.

T_2/T_1	t_c (hashing)	purity	ratio
2.0	0.497	0.881	1.77
1.0	0.291	0.549	1.89
0.5	0.187	0.347	1.85
0.2	0.105	0.186	1.77

The bracket also resolves the shape of the transition. Let $\boldsymbol{p}_n$ depend on n, with i.i.d. errors at each n, and set $x_n = \sqrt{n}(1 - H_n)/\sigma_n$.

Corollary 1. *If $\boldsymbol{p}_n \to \boldsymbol{p}_*$ with $H(\boldsymbol{p}_*) = 1$ and $\sigma(\boldsymbol{p}_*) > 0$, then for every pair of independent exact 2-design ensembles, $\overline{T}_n = \Phi(x_n) + O(\log n/\sqrt{n})$, where Φ is the standard normal distribution function.*

Proof. The Berry–Esseen theorem [19] gives $\mathbb{P}[s < n \mp a] = \Phi(x_n \mp a/\sqrt{n}\sigma_n) + O(n^{-1/2})$, uniformly because the third moment of the surprisal is bounded on a four-letter alphabet and σ_n is bounded away from zero. Insert into (4) with $a = 2\log_2 n$. □

The variance condition is necessary. Complete dephasing has $H = 1$ but $\sigma = 0$. The union of $d + 1$ mutually unbiased bases is an exact 2-design [20], and with the computational basis among them, $\overline{T}_{\mathrm{MUB}} = (d-1)(2d+1)/[d(d+1)^2] \to 0$, whereas $\overline{T}_{\mathrm{Haar}} = (d-1)/(2d-1) \to 1/2$. Two ensembles with the same two moments reach different critical limits, sharpening an observation of Ref. [8].

Memory lifetime.—An idling qubit with relaxation time T_1 and dephasing time $T_2 \le 2T_1$ is described, in the Pauli-twirl approximation standard in threshold analyses [21], by $p_x = p_y = (1 - e^{-t/T_1})/4$ and $p_z = (1 - e^{-t/T_2})/2 - p_x$. The scrambled memory retains its information until $H(t_c) = 1$. For $T_2 = T_1$ the noise is depolarizing, $r(t) = 1 - e^{-t/T_1}$, so

$$t_c = -\ln(1 - r_c)\,T_1 = 0.291\,T_1, \tag{9}$$

whereas purity certifies loss only after $(\ln 3/2)T_1 = 0.549\,T_1$. Across $0.2 \le T_2/T_1 \le 2$, the purity estimate exceeds t_c by a factor 1.77 to 1.89 (Table I). For pure dephasing ($T_1 \to \infty$), $p_z < 1/2$ and hence $H < 1$ at every finite t: the memory never loses its information, although for each fixed t the convergence rate of $\overline{T}_n \to 1$ decreases as t grows. Purity certifies nothing at all in this case, because $2(p_0^2 + p_z^2) \ge 1$. These statements are theorems for the twirled channel; for the untwirled amplitude-damping channel, which is not unital, they are approximations.

Biased noise shows a sharper contrast. Let the total error probability be p and the bias $\eta = p_z/(p_x + p_y)$. The hashing threshold rises with bias, from $p_c = 0.189$

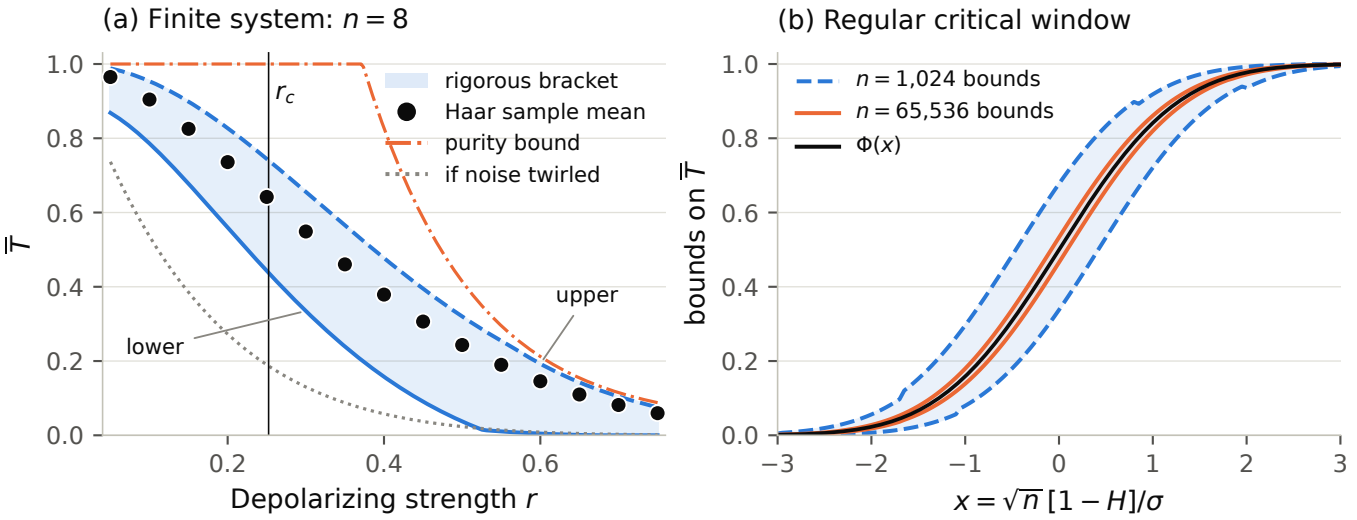


FIG. 1. Distinguishability after one layer of local depolarizing noise. (a) At $n = 8$, the shaded band lies between the projector lower bound (solid) and the two-class upper bound (dashed) of Eqs. (5) and (6), optimized over error-weight cutoffs; the thin vertical line marks r_c. Points are averages over 64 independent Haar-state pairs from full density matrices, with one-standard-error bars; the same pairs are used at every r. The dash-dotted curve is the purity bound, which drops below one only for $r \gtrsim 0.37$ at this size and certifies decay only above 0.423 asymptotically. The dotted gray curve is the depolarizing parameter θ reached if a further 2-design unitary twirled the noise; it equals the twirled distance up to a factor $1 - O(1/d)$. (b) The bracket (4) with $a = 2\log_2 n$, evaluated from binomial distributions at fixed $x = \sqrt{n}(1 - H)/\sigma$ for $n = 1024$ and 65536, approaches $\Phi(x)$ (black).

at $\eta = 1/2$ (depolarizing) to 0.278 at $\eta = 10$ and 0.390 at $\eta = 100$. Purity, by contrast, certifies loss only on a bounded window of p: for $\eta = 10$ only $0.387 < p < 0.705$, and for $\eta = 100$ only $0.455 < p < 0.555$. Strong biased noise concentrates on few Pauli errors, which raises the purity, so purity bounds fail precisely where biased-noise hardware [22] operates.

Information capacity of scrambled codebooks.—A memory that stores one of M messages in states $\psi_1, \ldots, \psi_M$, drawn independently from a state 2-design, is a classical code. Its rate is set by the same spectrum.

Theorem 2. *With the square-root measurement [23], the average error probability satisfies* $\mathbb{E}P_{\text{err}} \le 2\mathbb{P}[s \ge n - a] + 4(M-1)2^{-a}$*; this needs only a 1-design. For every measurement and every* $\gamma, a > 0$*, the average success probability satisfies*

$$\mathbb{E}P_{\text{succ}} \le \frac{\gamma}{M} + \frac{1}{4\gamma} + \frac{2^{-a}}{4} + \mathbb{P}[s < n - \log_2 \gamma + a], \quad (10)$$

which uses the second moments of the design.

Proof. For the first bound, take Π_i as above with $\mathcal{G} = \{s < n - a\}$ and apply the Hayashi–Nagaoka inequality $I - E_i \le 2(I - \Pi_i) + 4\sum_{j \ne i} \Pi_j$ [23]. Then $\operatorname{tr} \mathcal{N}(\psi_i)\Pi_i \ge p(\mathcal{G})$, and independence gives $\mathbb{E} \operatorname{tr} \mathcal{N}(\psi_i)\Pi_j = \operatorname{tr} \Pi_j/d < 2^{-a}$ for $j \ne i$. For the second, any POVM satisfies $\operatorname{tr} E_i \mathcal{N}(\psi_i) \le \operatorname{tr}(\mathcal{N}(\psi_i) - cI)_+ + c \operatorname{tr} E_i$, so $P_{\text{succ}} \le cd/M + M^{-1} \sum_i \operatorname{tr}(\mathcal{N}(\psi_i) - cI)_+$. Split $\mathcal{N}(\psi_i)$ at $p_P = c'$. The head contributes at most its mass. For the tail B, use $(\lambda - c)_+ \le \lambda^2/4c$ and $\mathbb{E} \operatorname{tr} B^2 = (dv_t + m_t^2)/(d+1) \le c' + 1/d$ from (3), then set $\gamma = cd$ and $c' = c2^{-a}$. □

For i.i.d. noise and $M = 2^{nR}$, the first bound vanishes for $R < 1 - H$, and Eq. (10) with $\gamma = 2^{n(R-\epsilon)}$ and $a = \epsilon n$ forces $\mathbb{E}P_{\text{succ}} \to 0$ for $R > \max(1 - H, 0)$. Scrambled codebooks therefore carry classical information at exactly the hashing rate $1 - H$, with a strong converse. The 2-design property is needed for the converse: computational-basis codewords, a 1-design, are blind to Z errors and can exceed $1 - H$. Product inputs with joint decoding achieve the capacity $1 - h_2(1 - r/2)$ of the depolarizing channel [24], which stays positive for all $r < 1$. Throughout $r_c < r < 1$, scrambling the encoding therefore costs the entire classical capacity; at r_c that capacity is 0.453 bits per qubit.

Error mitigation after one noisy layer.—Suppose a QEM protocol receives N copies of $\mathcal{N}(\rho)$ and must estimate $\operatorname{tr}(A\rho)$ to accuracy δ with probability at least $1 - \epsilon$, for all pure inputs and all observables with $\|A\| \le 1$. Two inputs whose expectation values differ by more than 2δ must then be distinguished from N copies, so $1 - 2\epsilon \le N \cdot \frac{1}{2}\|\mathcal{N}(\psi_1) - \mathcal{N}(\psi_2)\|_1$ [2]. By Markov's inequality, a random 2-design pair has distance at most $2\overline{T}$ with probability at least 1/2, and overlap $|\langle\psi_1|\psi_2\rangle|^2 \le 4/d$ with probability at least 3/4. Taking $A = \psi_1$ then gives

$$N \ge \frac{1 - 2\epsilon}{2\overline{T}_n} \quad (11)$$

whenever $2\delta < 1 - 4/d$. Beyond the hashing point,

$\overline{T}_n = e^{-\Omega(n)}$, so a single noisy layer already forces exponentially many samples. For one layer, Theorem 3 of Ref. [2] gives a bound that decreases with n, and the purity route of Ref. [6] applies only for $r > 0.423$. Below threshold, Eq. (11) gives no exponential bound. The bound concerns protocols that only receive copies of the noisy state and that work for global observables.

Discussion.—A scrambled encoding is not fragile under local Pauli noise. Its information survives until the error entropy reaches one bit per qubit, and this one threshold controls three quantities at once: how long a scrambled memory remembers, how much classical information a scrambled codebook carries, and when a single noisy layer makes error mitigation exponentially expensive. In each case, purity-based reasoning misplaces the boundary, by nearly a factor of two in memory lifetime and by an unbounded amount for biased noise and pure dephasing. The reason is the same throughout: purity weighs the few likely errors like the many unlikely ones, whereas the optimal measurement resolves the likely ones.

The bracket rests only on the error distribution and the dimension, so it holds for every 2-design and for correlated noise. For correlated errors, the full information spectrum replaces the entropy: global depolarization has entropy rate $2(1-\theta) > 1$ for $\theta < 1/2$, yet $\overline{T} \to \theta$, which the bracket reproduces. The main open problem is depth. In deeper circuits, later scrambling layers partly twirl earlier noise toward global depolarization. Tracking the information spectrum layer by layer would decide whether a finite threshold survives at depth. Determining Haar decay rates away from threshold, where random stabilizer states, a 3-design, decay faster than the achievable exponent [13], is a second open question.

Acknowledgments.—Generative AI was used substantively in this work. Claude Opus 5.5 (Anthropic), accessed through Claude Code in October 2026, with review passes by other Claude model instances, was directed by the author to select the problem, draft and revise the derivations, survey the literature, write the analysis code and the Lean formalization, generate the figures and table, and draft the text. A review with an OpenAI GPT model [version to be specified by the author] identified the omitted Ref. [8]. The author is responsible for all content. Proofs were rederived independently, Theorem 1 and Corollary 1 were machine-checked in Lean as described in the text, the exact counterexamples were checked by explicit enumeration for $n \leq 2$, the codebook bounds were checked against semidefinite-programming optima for $n = 2$, numerical claims were checked against independent implementations recorded with the data, and cited statements were checked against the primary sources.

Data availability.—The data that support the findings of this article, the code that generates every figure, table, and numerical check, and the Lean formalization accompany this submission and will be deposited in a public repository [13].

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